# Lab 7 – Article Review: Experiment

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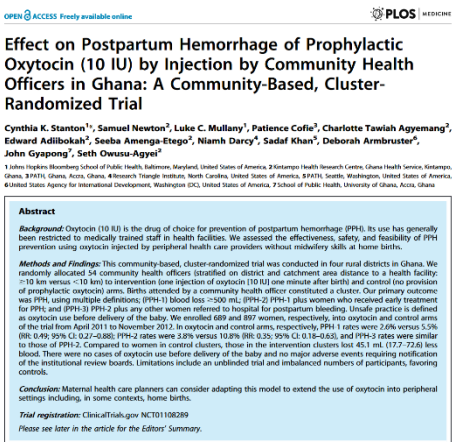
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## Formatting Requirements

- Please submit your lab report as a **pdf** to Gradescope.
- When you upload to Gradescope, please **match pages** with the **question number**.
- Be sure that all **group members** are **added** in your submission to Gradescope (click view/edit group on the top right of the page once shown your final submission after matching pages).

## Assignment Overview

- In this assignment, you will be reading and summarizing key points from the article titled: "Effect on Postpartum Hemorrhage of Prophylactic Oxytocin (10 IU) by Injection by Community Health Officers in Ghana: A Community-Based, Cluster-Randomized Trial."
- The goal of this lab is to identify the aims of this study, the design, the statistical results, and the claims they are making from those results.



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**Effect on Postpartum Hemorrhage of Prophylactic Oxytocin (10 IU) by Injection by Community Health Officers in Ghana: A Community-Based, Cluster-Randomized Trial**

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**Abstract**

***Background:*** Oxytocin (10 IU) is the drug of choice for prevention of postpartum hemorrhage (PPH). Its use has generally been restricted to medically trained staff in health facilities. We assessed the effectiveness, safety, and feasibility of PPH prevention using oxytocin injected by peripheral health care providers without midwifery skills at home births.

***Methods and Findings:*** This community-based, cluster-randomized trial was conducted in four rural districts in Ghana. We randomly allocated 54 community health officers (stratified on district and catchment area distance to a health facility: ≥10 km versus <10 km) to intervention (one injection of oxytocin [10 IU] one minute after birth) and control (no provision of prophylactic oxytocin) arms. Births attended by a community health officer constituted a cluster. Our primary outcome was PPH, using multiple definitions; (PPH-1) blood loss >500 mL; (PPH-2) PPH-1 plus women who received early treatment for PPH; and (PPH-3) PPH-2 plus any other women referred to hospital for postpartum bleeding. Unsafe practice is defined as oxytocin use before delivery of the baby. We enrolled 689 and 897 women, respectively, into oxytocin and control arms of the trial from April 2011 to November 2012. In oxytocin and control arms, respectively, PPH 1 rates were 2.6% versus 5.5% (RR: 0.49; 95% CI: 0.27–0.88); PPH-2 rates were 3.8% versus 10.8% (RR: 0.35; 95% CI: 0.18–0.65), and PPH-3 rates were similar to those of PPH-2. Compared to women in control clusters, those in the intervention clusters lost 45.1 mL (17.7–72.6) less blood. There were no cases of oxytocin use before delivery of the baby and no major adverse events requiring notification of the institutional review boards. Limitations include an unblinded trial and imbalanced numbers of participants, favoring controls.

***Conclusion:*** Maternal health care planners can consider adapting this model to extend the use of oxytocin into peripheral settings including, in some contexts, home births.

***Trial registration:*** ClinicalTrials.gov NCT01108289

*Please see later in the article for the Editors' Summary.*

## Tips for reading research articles

- You won't understand a lot of what is being said in this article, and that's ok! Focus instead on making sense of the study's primary aims, the general design, and the contribution
- Whenever you see a term used multiple times, but aren't sure what it is, take a few seconds and search it online!
- **Abstracts** are great at helping you pull out key details. You should read this first, then at various stages of reading the rest of the paper, come back and read it again!
- Once you have finished working on all questions, come back to the beginning and revise/enhance your answers based on the new knowledge you gained later!

## Acknowledgment

- Thanks to Laura Le and others at the University of Minnesota School of Public Health! This lab is an adaption from an assignment that they wrote and kindly shared with me.

**Read the Abstract and the Introduction sections on pages 1-2.** *(If needed, you may also check out the Editor's Summary on the last page!)*

**Question 1 (6pts)**: Briefly discuss the **aims** of this study.

a) What is the response variable? *(suggested 1-2 sentences)*

b) What were the treatment/control factors being compared? *(suggested 1-2 sentences)*

c) In your own words, what question are the researchers trying to answer with this study?

d) *Why* was this study conducted? What is the potential significance of this study's findings? *(suggested 2-3 sentences)*

**Answers:**

a) The response variable in this study is the incidence of postpartum hemorrhage (PPH), includes PPH-1, PPH-2 and PPH-3.

b) The study compared the use of prophylactic oxytocin (10 IU) administered by community health officers (treatment) with the absence of prophylactic oxytocin (control) during home births.

c) The researchers are attempting to see whether prophylactic oxytocin administration by community health officials, who may lack midwifery expertise, is beneficial in reducing postpartum hemorrhage during births at home in low-income areas.

d) The purpose of this study was to address the serious problem of maternal mortality owing to postpartum hemorrhage in low-income countries, notably in Sub-Saharan Africa. The findings of this study have the potential to be significant since they provide information on the effectiveness, safety, and feasibility of administering oxytocin to prevent PPH during home births by inexperienced attendants. If effective, it could lead to the increased use of oxytocin in outlying areas and enhance maternal health outcomes in areas with limited access to professional birth attendants.

**Skim the "Methods" section (especially the first couple sections)**

**Question 2 (4pts):** Briefly discuss the experimental design of this study

a) What does it mean when the authors say they used "cluster randomization" in this study? What constitutes a "cluster," and how would we distinguish that from the unit of observation in this study? *(suggested 2-3 sentences)*

b) *Why* did the researchers choose to randomize at the cluster level rather than at the unit of observation level? *(suggested 2-3 sentences)*

**Answers:**

a) "Cluster randomization" means that the random allocation of the treatment (oxytocin) or control (no oxytocin) was done at the level of clusters. A cluster in this context refers to a group of community health officers (CHOs) who were responsible for attending births. The unit of observation in this study was the individual women giving birth. Cluster randomization means that all births attended by a specific CHO (cluster) received the same treatment, and the outcomes of multiple births within that cluster were analyzed together.

b) For simpler study implementation, the researchers chose to randomize at the cluster level rather than at the individual woman level. Individual randomization would have meant that each woman may have received a different therapy within the same CHO's catchment area, making consistent implementation difficult. They guaranteed that each CHO was allocated a consistent set of tasks by randomizing at the cluster level, which expedited the training and implementation process and minimized the risk of unintentional variances in the intervention's use.

**Question 3 (6pts):** Let's consider the study's internal validity (ability to draw causal claims) regarding the effects of oxytocin

a) Focus first on Table 1, which compares the 682 women who received oxytocin versus the 887 women in the control group. Would you say that these groups are quite similar and balanced, or are there any systematic demographic differences that stand out to you?

b) Focus next on Table 2. Would you say that the procedures followed for births under each condition are similar and balanced, or are there any systematic differences in the situations/protocols followed under each condition that stand out to you?

c) On page 12 in the first full paragraph, the authors address some possible differences. Do either of these differences seem to threaten the effects found from oxytocin?

**Answers:**

a) The study demonstrates good internal validity regarding the comparability of the oxytocin and control groups with respect to demographic characteristics. This balance enhances the study's ability to draw causal claims about the effects of oxytocin on postpartum hemorrhage without being significantly confounded by demographic differences.
b) The recruitment/enrollment techniques and measurement indicators demonstrate a well balanced approach between the two groups. In the control group, however, there are systematic disparities in the timing and practice of drape insertion, prophylactic injection delivery, and timing. These variations may have an impact on the study's internal validity, particularly when comparing the outcomes associated to oxytocin prophylaxis. When evaluating the benefits of oxytocin on avoiding postpartum hemorrhage, these differences must be considered.
c) The differences in CHO behaviors are not likely to threaten the effects observed with oxytocin. The delay in unfolding the drape for blood-loss measurement in the control group might have underestimated blood loss in that group, but it does not compromise the effectiveness of oxytocin. Additionally, the more frequent referral of women prior to delivery by CHOs in the oxytocin group did not indicate a higher risk of postpartum hemorrhage, so it does not appear to affect the observed positive effects of oxytocin in preventing hemorrhage.

**Look at Table 3**

**Question 4 (5pts):** What might we learn from Table 3?

a) Briefly explain how the values in the percentage column are being calculated. *(suggested 1-2 sentences)*

b) The value 1.000 in this table is actually a typo! Based on the data reported in this table, what should that number actually be?

c) For which outcome comparisons are we confident in concluding that oxytocin is likely reducing the risk for? What might you use as evidence to make that claim? Note: there is no *objective* benchmark for making this decision, so just be clear how *you* are making that decision! *(suggested 2-3 sentences)*

**Answers:**

a) The values are calculated by dividing the number of cases (N) for each outcome by the total number of women in the group (either control or oxytocin) and then multiplying by 100 to express it as a percentage.
b) This value indicates a relative risk. However, the relative risk for Severe PPH should be 0.1, not 1.000.
c) We can be confident in concluding that oxytocin is likely reducing the risk for PPH-1, PPH-2, and PPH-3. This conclusion is supported by the relative risks for these outcomes being significantly less than 1 (0.491, 0.338, 0.361) and the associated p-values (0.02, 0.001, 0.001) indicating statistical significance.

**Look at Table 4**

**Question 5 (4pts):** What might we learn from Table 4?

a) How might you interpret what the value 1.309 is representing in context?

b) Would you say there is strong evidence that oxytocin is likely increasing the risk for stillbirth or any other adverse effect here?

**Answers:**

a) This value represents the relative risk of stillbirth in the oxytocin group compared to the control group. A relative risk of 1.309 suggests that the risk of stillbirth in the oxytocin group is 1.309 times higher than the risk in the control group.
b) There is no strong evidence to suggest that oxytocin is likely increasing the risk of stillbirth or any other adverse effect. Because the 95% confidence interval includes 1.000, this difference is not statistically significant.

**Question 6 (6pts):** In biomedical research, the "number needed to treat" (NNT) measures how many people would need to be treated before you would expect one adverse effect to be avoided. For example, if the absolute risk in the treatment group is 1% (0.01) and the absolute risk in the comparison group is 3% (0.03), then the number needed to treat is 1 / (0.03 – 0.01) = 1/0.02 = 50.

a) In Table 3, what is the NNT before we expect to ***prevent*** one case of **PPH-3** as a result of oxytocin?

b) In Table 4, we are instead focusing on possible adverse effects of oxytocin. What is the NNT before we expect to ***inflict*** one case of **stillbirth** as a result of oxytocin?

c) How might you compare the benefit of oxytocin in preventing PPH-3 to the risk of oxytocin in inflicting a stillbirth? There is no *objective* answer to this question, but you should use the statistical results as an important consideration in making this risk-benefit comparison!

**Answers:**

a) The percentage of PPH-3 in the control group is 11.1%. In the oxytocin group, it's 4.1%. The absolute risk reduction is 11.1% - 4.1% = 7%. Therefore, the NNT is 1 / 0.07, which is approximately 14.29.
b) In Table 4, the percentage of stillbirth in the control group is 1.6%. In the oxytocin group, it is 2.1%. The absolute risk increase is 2.1% - 1.6% = 0.5%. Therefore, the NNT for inflicting one stillbirth due to oxytocin is 1 / 0.005, which is 200.
c) Comparing the benefit of oxytocin in preventing PPH-3 to the risk of oxytocin in inflicting a stillbirth, we find that the NNT for preventing one case of PPH-3 is much smaller (14.29) compared to the NNT for inflicting one case of stillbirth (200). This indicates that the benefit of oxytocin in preventing PPH-3 outweighs the risk of inflicting a stillbirth.

**Read the last 4 paragraphs of the "Discussion" section**

**Question 7 (4pts):** Briefly describe the contributions this article made.

a) Briefly summarize the primary finding of this article. What "answer" might they give to the research question you wrote up in Question 1?

b) Why might the findings of this study be limited in practice for birth care to the broader population being targeted?

**Answers:**

a) The administration of intramuscular prophylactic oxytocin by peripheral health care providers without midwifery skills effectively reduces the risk of postpartum hemorrhage at home births in research conditions. This study would answer that prophylactic oxytocin is an effective intervention for reducing PPH during home births.

b) This is because the findings were obtained under research conditions, which may not fully replicate the real-world challenges and variations in healthcare settings. The study's success relied on the compliance and availability of CHOs, which may not be feasible in all areas. Additionally, the cost and availability of oxytocin in Uniject can be limiting factors for its widespread implementation.